

Project Brief — Final

TEAM SATURN

First Commit Hackathon · Bharat Builds Tour Stop 01 · 17–20 September 2026 · Polaris School of Technology, Bengaluru (in-person day 19 Sept) + online

Team of 3 · Track: Build It · Auto-considered for Ship It and Best UI

■ Share what your meters learn. Not what your meters see.

Contents

Part I — Where this came from 1. The origin problem · 2. The resource gap · 3. Why raw data can't be pooled · 4. Why it fits this hackathon

Part II — Research foundation 5. Primary sources · 6. The 2026 landscape · 7. What DISCOMs and regulators say they want

Part III — The system 8. What we are building · 9. Architecture · 10. AWS / open-source stack · 11. Data

Part IV — Execution 12. Build plan · 13. Team split · 14. Demo video · 15. Judging map · 16. Risks · 17. Pre-Thursday checklist

Part V — Beyond the weekend 18. Research path · 19. Appendix — the pitch at three lengths

PART I — WHERE THIS CAME FROM

1. The origin problem — power theft in India

India loses roughly ■1.32 lakh crore (about \$16 billion) every year to electricity theft, the single largest driver of AT&C losses across the country's distribution companies (DISCOMs) and the highest theft-driven loss of any country. Nationally, AT&C losses fell to a record low of 15.04% in FY 2024–25 — real progress, but still meaning a meaningful share of every unit purchased is never recovered.

The picture is deeply uneven. The gap between the best- and worst-performing DISCOMs runs to about 45.8 percentage points. Well-instrumented private operators such as Delhi's TPDDL (5.91%) and BRPL (6.58%) sit far below the national average, while under-resourced regions, particularly in the Northeast, remain far above it.

2. The resource gap

The DISCOMs that perform well aren't smarter about theft — they're better resourced. Adani Electricity alone ran 36,720 mass raids and registered 486 FIRs in FY 2025–26 to bring its losses down to 4.46%. A small or rural DISCOM does not have that scale of manpower, meters, or analytics staff, and it sees too few theft cases in isolation to learn a reliable pattern on its own.

The resource gap, not a skill gap, is what we're building against. Everything since follows from that.

3. Why raw data can't be pooled

The obvious fix — have every DISCOM share its consumption data with every other — doesn't happen and largely can't. Household-level consumption data is sensitive: it exposes occupancy, business activity, and income proxies. DISCOMs are separate legal entities, mostly state-owned, with no shared regulatory pathway or incentive to hand raw consumer records to a neighbouring utility. So each substation re-learns theft patterns from first principles, even where the underlying signature — a load drop at a billing boundary, a tamper-consistent shape, a mismatch between declared connection type and usage — generalises reasonably well across regions.

4. Why it fits this hackathon

The theme is open — a real problem, in any sector, that matters to you, your community, or the country. Power theft costs roughly 2.4% of GDP and touches every honest, paying consumer through cross-subsidy. It is concretely buildable on the open-source AWS stack in four days, and it gives us a measurable before/after result rather than a described architecture.

PART II — RESEARCH FOUNDATION

5. Primary sources — verified

Source	Link
SGCC Electricity Theft Detection Dataset — Zheng, Yang, Niu, Dai, Zhou, “Wide and Deep Convolutional Neural Networks for Electricity-Theft Detection to Secure Smart Grids,” IEEE Trans. Industrial Informatics. Real consumption data, 42,372 consumers, 1,035 days.	kaggle.com/datasets/bensalem14/sgcc-dataset
EnsembleNTLDetect — Kulkarni et al., arXiv:2110.04502. Data pre-processing and ensemble learning framework for non-technical-loss (theft) detection on the SGCC dataset.	arxiv.org/abs/2110.04502
Electricity Theft Detection with Self-Attention — Finardi et al., arXiv:2002.06219. Sequence-model approach to the same detection problem; useful baseline reference.	arxiv.org/pdf/2002.06219
Combined MIC + CFSFDP approach — Tsinghua University (State Key Lab of Power Systems). Correlation- and clustering-based detection without requiring extra system information — close to what a signature-sharing approach would need.	arxiv.org/html/2411.06649v1

6. The 2026 landscape

AT&C losses have fallen from 27.34% in 2008–09 to 15.04% in 2024–25, the lowest on record, and India's DISCOMs posted a collective profit for the first time in over a decade (■2,701 crore in FY 2024–25) — a reversal against accumulated losses of ■6.47 lakh crore. Nearly half of DISCOMs still sit above 15% losses,

and the gap between the cost of supply and revenue realised remains wide in the weaker states. Peak power demand hit a record ~271 GW in May 2026, adding pressure on a grid that is still absorbing these losses.

7. What DISCOMs and regulators say they want

The Revamped Distribution Sector Scheme (RDSS) targets AT&C losses of 12–15% nationally and is funding the rollout of smart meters — 59.7 million installed so far — alongside stronger billing and collection discipline. The stated bottleneck across every source we found is execution and DISCOM-level granular accountability, not a lack of ambition. A system that lets under-resourced DISCOMs benefit from patterns learned elsewhere, without a data-sharing mandate, sits directly inside that gap.

PART III — THE SYSTEM

8. What we are building

A theft-anomaly detection system where every simulated substation trains its own local model, but instead of staying siloed, each site can contribute an **abstracted anomaly signature** — the shape of what theft looks like, with no consumer ID, no raw time series, no location — to a shared pattern library. A policy engine (Cedar) enforces, provably, that only abstracted signatures can leave a site; raw consumption records are rejected by rule, not by promise.

9. Architecture

Per-site layer: a lightweight anomaly detector (isolation forest / simple classifier on load-shape features) trained only on that site's own data — this is the isolated baseline, and it is deliberately weak on low-data sites.

Abstraction layer: each detected anomaly is converted into a signature vector describing its shape (e.g. magnitude of load drop, timing relative to billing cycle, recovery time) with all identifying fields stripped.

Policy layer: Cedar policies gate what leaves a site — deny anything carrying a meter ID, raw series, or household field; permit only signature vectors below a defined granularity. This is our block-then-pass proof.

Shared layer: permitted signatures land in a shared pattern store. Each site checks new consumption against its own model *and* the pooled library.

10. AWS / open-source stack

Build It track, no AWS account required: **Strands Agents SDK** orchestrates each site's detect → abstract → request-to-share flow; **Cedar** enforces the sharing policy; **SAM CLI + LocalStack** gives us a serverless-shaped backend on localhost; **OpenSearch** indexes flagged anomalies across sites for the dashboard.

11. Data

The SGCC dataset (42,372 real consumers, theft-labelled) is split into 3–4 synthetic “sites” with different consumer mixes, simulating siloed substations without needing real DISCOM access.

PART IV — EXECUTION

12. Build plan

Thursday: data split into synthetic sites, local anomaly detector running end to end on one site, Cedar hello-world (trivial permit/forbid) working.

Friday: signature extraction pipeline; Cedar policy gating raw vs. abstracted records; block-then-pass test passing; first isolated-vs-pooled numbers on one low-data site.

Saturday: dashboard and shared pattern store; feature freeze at night; video storyboard finalised.

Sunday: record demo video, polish the before/after result, submit in the afternoon — not at the deadline.

13. Team split

Data + model: data split, local anomaly detector, and the isolated-vs-pooled experiment — this produces the one number the judges actually score, so protect this person's time.

Policy + pipeline: Cedar policies, signature extraction, shared pattern store.

Demo + video: dashboard showing flagged anomalies across sites and the block-then-pass proof live; owns the three-minute video.

14. Demo video

Three minutes, recorded — no live demo, so this is what judges actually see. Structure: the problem (45.8-point gap between DISCOMs) → the boundary problem (why raw data can't be pooled) → the system in action (block-then-pass, then a site's detection improving after receiving shared signatures) → the honest caveat (validated on a public dataset split into synthetic sites, not live meter data) → what AWS/Cedar made possible.

15. Judging map

Criterion	How this project answers it
Idea and Impact	Power theft costs ■1.32 lakh crore/year and 2.4% of GDP; the resource gap between DISCOMs is real and documented, not assumed.
Built on AWS	Strands orchestrates the flow, Cedar enforces the policy boundary — mandatory, not decorative, to the idea working at all.
Learning	None of us has built a policy-enforced data-sharing system before; Cedar and the signature-abstraction design are both new ground this weekend.
Execution	One working feature — the block-then-pass proof plus one measured before/after number — beats five described capabilities.
The demo video	Leads with the real stakes, shows the boundary being enforced live, and states plainly what is proven versus what is future work.

16. Risks

Risk	Mitigation
Cedar syntax eats a day	Hello-world permit/forbid running before Thursday. If still fighting Friday noon, fall back to a rule-based gate and say so honestly in the write-up.
Overbuilding	Hard rule: one detector, one policy set, one before/after number beats a full federated-learning system that half-works. Feature freeze Saturday night.
Result doesn't move	Test the isolated-vs-pooled comparison Friday, not Saturday night — leaves time to adjust the signature design if the numbers are flat.
Video rushed	Storyboard before Thursday, practice recording Saturday night.
Submission fails	Submit Sunday afternoon, not at the deadline.
"Isn't this just theft detection with extra steps?"	No — ordinary theft detection stays siloed per site. This moves generalised patterns, blocks raw records, and the boundary is machine-enforced and auditable. Have the answer ready for mentors.
"Why not real federated learning?"	A shared signature library is a simpler, buildable-in-four-days version of the same idea; full federated weight-averaging is the documented next step, not this weekend's scope.

17. Pre-Thursday checklist

All three of us. Nothing here is building — it is removing Thursday friction.

- WeMakeDevs hackathon check-in complete — **each person**
- AWS Builder Center profile created — **each person**
- Student status verified — each person. This gates scoring, not just entry
- Joined WeMakeDevs Discord and the Builder Center space
- Strands installed, hello-world agent running — each person
- Cedar installed, one trivial permit/forbid evaluating — each person
- SGCC dataset downloaded and split into 3–4 synthetic sites
- New GitHub repo created, empty, all three with push access
- Scope agreed in writing: N sites, one signature type, one policy set
- Video storyboard sketched on paper
- Project name decided

PART V — BEYOND THE WEEKEND

18. Research path after the hackathon

This weekend's build is the scoped version. The open question worth publishing: which theft signatures transfer across regions with different consumer mixes, and which don't? That is the same question our robotics-domain draft asked about warehouse morphologies, now grounded in a real, publicly documented Indian infrastructure problem.

Near term: benchmark the shared-signature approach against a proper isolated baseline on the full SGCC dataset, not just a synthetic split, and study where a signature learned on one consumer mix stops generalising to another.

Medium term: differential privacy on the signatures themselves, with a measured privacy–accuracy curve — be precise that abstraction alone is a real privacy improvement, not a formal guarantee, since re-identification attacks against aggregated statistics do exist.

Routes: a smart-grid or applied-ML workshop is a realistic venue for this at the synthetic-data stage; a real DISCOM pilot is the credible next step if the signature-transfer result holds up.

19. Appendix — the pitch at three lengths

One line: Share what your meters learn, not what your meters see.

Thirty seconds: India loses ■1.32 lakh crore a year to power theft, and the gap between the best- and worst-performing DISCOMs is 45.8 percentage points — not because smaller DISCOMs are worse at their jobs, but because they see too few theft cases to learn the pattern alone, and raw consumption data is too sensitive to pool across utilities. We built a system where sites share the abstracted shape of a theft pattern, never the raw record, with the boundary enforced by policy, not promise — and showed that a low-data site catches more theft once it can see what other sites have learned.

Two minutes: Walk through the origin (the resource gap between DISCOMs), the constraint (why raw data can't move), the mechanism (per-site detection, abstracted signatures, Cedar-enforced sharing), and the proof (the block-then-pass demo plus the isolated-vs-pooled number) — closing on the honest caveat that this is validated on a public dataset split into synthetic sites, with a real DISCOM pilot as the open next step.